PRACTICAL EXERCISE 30

Developing critical research skills by auditing AI-generated claims, identifying hallucinated sources, and correcting inaccuracies.

USER PROMPT

"Explain five important facts about Generative AI and provide reliable sources that can be used to verify each claim."

- **Task 1:** Identify five factual claims in the response.
- **Task 2:** Locate reliable sources for each claim independently.
- **Task 3:** Check whether the provided sources actually support the claims.
- **Task 4:** Correct inaccurate or unsupported information.
- **Task 5:** Submit the original response and the verified version.

PHASE 1: ORIGINAL AI OUTPUT (UNVERIFIED)

Five Facts About Generative AI

1. Generative AI models like GANs (Generative Adversarial Networks) pit two neural networks against each other to create novel content.
   Source: Goodfellow et al., "Generative Adversarial Nets" (2014)
2. ChatGPT became the fastest-growing consumer application in history, reaching 100 million active users in just two months.
   Source: UBS Research / Reuters Technology Report (2023)
3. Large Language Models (LLMs) hallucinate because their core mechanism is probabilistically predicting the next likely word, not querying a database of absolute facts.
   Source: IBM Research Guide on AI Architecture
4. The core neural architecture behind modern Generative AI like GPT is continuously trained on live internet data in real-time to stay perfectly up to date.
   Source: OpenAI 2023 Dynamic Architecture Whitepaper
5. Under US Law, any image generated by an AI is automatically granted full copyright protection as long as a human authored the text prompt.
   Source: US Copyright Office Official Ruling on AI Art (2023)

PHASE 2: STUDENT FACT-CHECKING MATRIX

Claim	Student Research Findings	Source Check	Verdict
1 (GANs)	The description of two neural networks (generator and discriminator) competing is mechanically accurate.	Source exists and matches claim.	✅ **Verified True**
2 (100M)	Analytics data corroborates that ChatGPT reached 100M users in January 2023, two months after launch.	Source exists and matches claim.	✅ **Verified True**
3 (Prediction)	LLMs function as next-token predictors. Hallucination stems from statistical probability lacking real-world grounding.	Source concept is accurate.	✅ **Verified True**
4 (Real-time)	LLMs have training cutoff dates. While some tools can search the web dynamically, the core weights are NOT continuously trained in real-time.	**Hallucinated Source:** No such whitepaper exists.	❌ **False / Hallucination**
5 (Copyright)	The USCO ruled that AI-generated images lack human authorship. Prompting alone does not qualify for copyright.	**Fabricated Context:** The USCO ruled the exact opposite.	❌ **False / Reversed Fact**

PHASE 3: VERIFIED & CORRECTED OUTPUT

Corrections Applied to Original Output:

Corrected Fact 4: The core neural architecture behind LLMs like GPT has a specific training data cutoff date. While models can be integrated with external search tools to retrieve current information, their underlying weights are not continuously trained in real-time.

Corrected Fact 5: Under current US Law, images generated entirely by AI are generally *not* eligible for copyright protection, as the US Copyright Office requires a sufficient amount of human authorship. Simply writing a text prompt is currently insufficient to claim copyright over the resulting image.

Key Learning Outcomes

1. The Danger of Plausible Falsehoods: AI models generate text that sounds highly authoritative. Fact 4 and Fact 5 sound completely legitimate, making them dangerous if relied upon without verification.
2. Source Hallucination: Models will invent academic papers, whitepapers, or legal rulings (like the "OpenAI 2023 Dynamic Architecture Whitepaper") to satisfy a prompt requesting citations.
3. Human Accountability: The ultimate responsibility for truth lies with the human operator. AI is a drafting and ideation tool, not an infallible search engine or legal advisor.